Name
Date
Course

# Graphing Quadratic Expressions (Part VII)

## The Three Forms of a Quadratic Expression

| Standard Form | Intercept Form | Vertex Form |
|---|---|---|
| $y = ax^2 + bx + c$ | $y = a(x-m)(x-n)$ | $y = a(x-h)^2 + k$ |

## Tools for Graphing Quadratic Expressions

### Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

### Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x-value of the vertex is $\frac{m+n}{2}$.

* Intercept form is not commonly used to graph a quadratic expression if the corresponding graph does not move through the x-axis.

### Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then $(h,k)$ is the vertex.

### Finding the Y-intercept in Standard Form

If $y = ax^2 + bx + c$ is graphed on an xy-plane, then $(0,c)$ is the y-intercept of the graph.

## The Focus and Directrix

### Finding the Focus and Directrix

If a quadratic expression is written in any of the forms $y = ax^2 + bx + c$, $y = a(x-m)(x-n)$, or $y = a(x-h)^2 + k$, and $(V_x, V_y)$ are the coordinates of the vertex, then the coordinates of the focus are $(V_x, V_y + P)$ and the equation of the directrix is $y = V_y - P$ if P is defined to be the expression below.

$$P = \frac{1}{4a}$$

# Summary for Graphing Quadratic Expressions

## Steps for Graphing a Parabola that Opens Up or Down

I) Identify the form of the quadratic expression and use it to plot the vertex of the parabola (Standard Form: $V_x = -b/2a$, Intercept Form: $V_x = \frac{m+n}{2}$, Vertex Form: $V(h,k)$).

II) Draw the line of symmetry (always a vertical line through the vertex) and write its equation (always $x = V_x$).

III) Let $x=0$ in the original expression and solve for $y$ to plot the y-intercept.

IV) Let $y=0$ in the original expression and solve for $x$ to plot the x-intercepts (be aware that there can be two, one, or zero x-intercepts).

a) If the expression is in intercept form, then you do not have to do anything; you can see the x-intercepts.

b) If the expression is in vertex form, then you must solve an equation with a square.

c) If the expression is in standard form, then there are three ways to find the x-intercepts: i) factoring, ii) completing the square, or iii) the quadratic formula. You should never complete the square unless a problem specifically asks you to because completing the square is the longest and most difficult method. You should always try to factor if you can, but if the expression is not easily factorable or you are not sure if it can be factored, use the quadratic formula.

V) Find P, which is $1/4a$ and use it to draw the directrix (always a horizontal line). Write its equation (always $y = V_y - P$).

VI) Plot the focus [always $(V_x, V_y + P)$].

Graph the following quadratic expressions. Plot the vertex, y-intercept, x-intercepts (if any), and focus. Draw the line of symmetry and the directrix and write their equations.

① $y = x^2 + 2x - 8$

$a = 1,\ b = 2,\ c = -8$

I) Vertex

$V_x = -\frac{b}{2a} = -\frac{2}{2(1)} = -1$

$V_y = (-1)^2 + 2(-1) - 8 = -9$

$V(-1, -9)$

II) Line of Symmetry

$x = -1$

III) Y-Intercept: $x = 0$

$(0, -8)$

IV) X-Intercepts: $y = 0$

$0 = x^2 + 2x - 8$

$0 = (x+4)(x-2)$

$x = -4$ or $x = 2$

$(-4, 0)$ & $(2, 0)$

V) Directrix

$P = \frac{1}{4a}$

$P = \frac{1}{4(1)}$

$P = \frac{1}{4}$

$y = V_y - P$

$y = -9 - \frac{1}{4}$

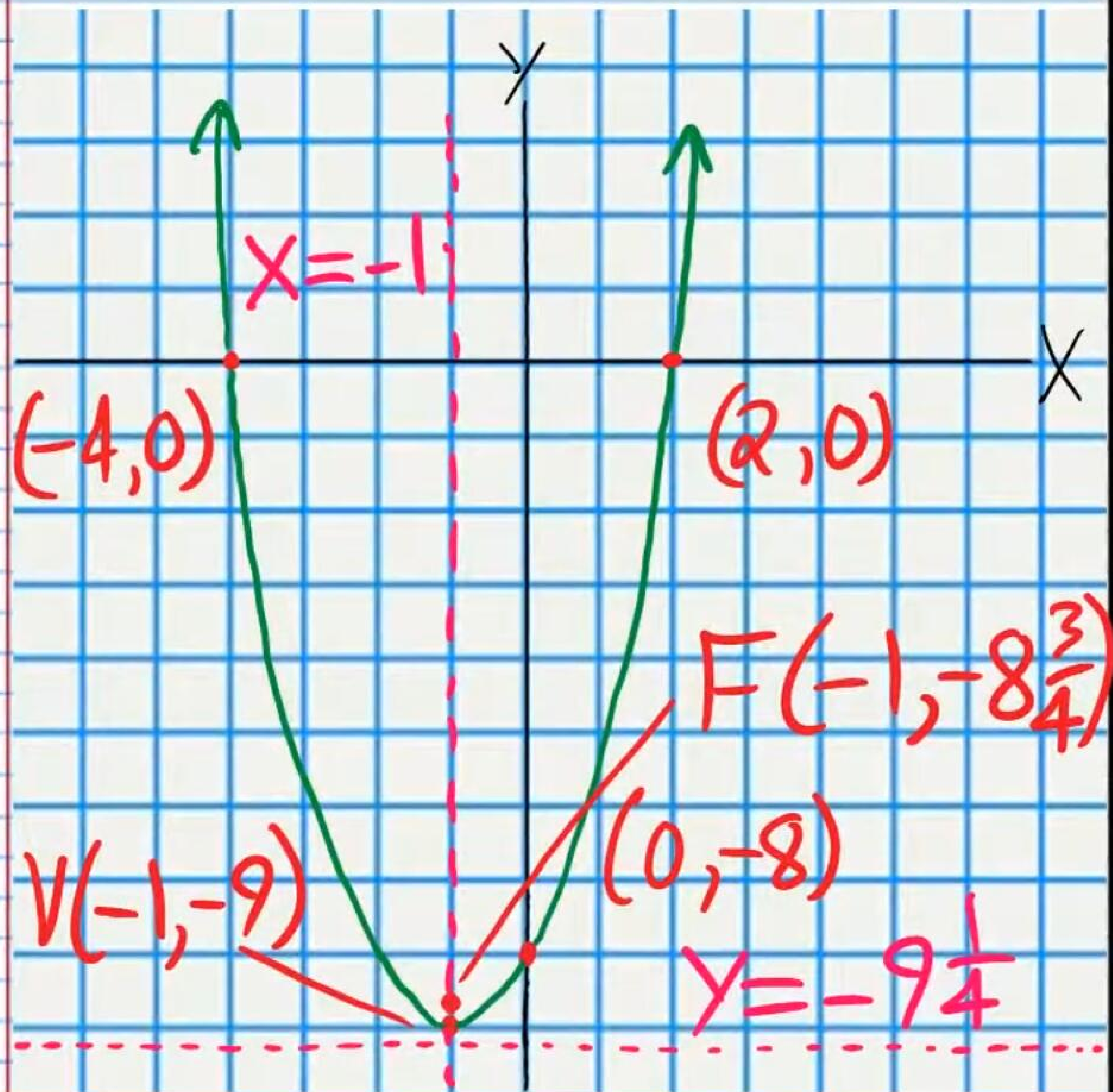
$y = -9\frac{1}{4}$

VI) Focus

$F(V_x, V_y + P)$

$F(-1, -9 + \frac{1}{4})$

$F(-1, -8\frac{3}{4})$



② $-\frac{1}{2}(x+3)(x-3)=y$

$a=-\frac{1}{2},\ m=-3,\ n=3$

I) Vertex

$V_x=\frac{m+n}{2}=\frac{-3+3}{2}=0$

$V_y=-\frac{1}{2}(0+3)(0-3)=4.5$

$V(0,4.5)$

II) Line of Symmetry

$x=0$

III) Y-Intercept: $x=0$

$(0,4.5)$

IV) X-Intercepts: $y=0$

$(-3,0)\ \&\ (3,0)$

V) Directrix

$P=\frac{1}{4a}=\frac{1}{4(-\frac{1}{2})}=-\frac{1}{2}$

$y=V_y-P$

$y=4.5-(-\frac{1}{2})$

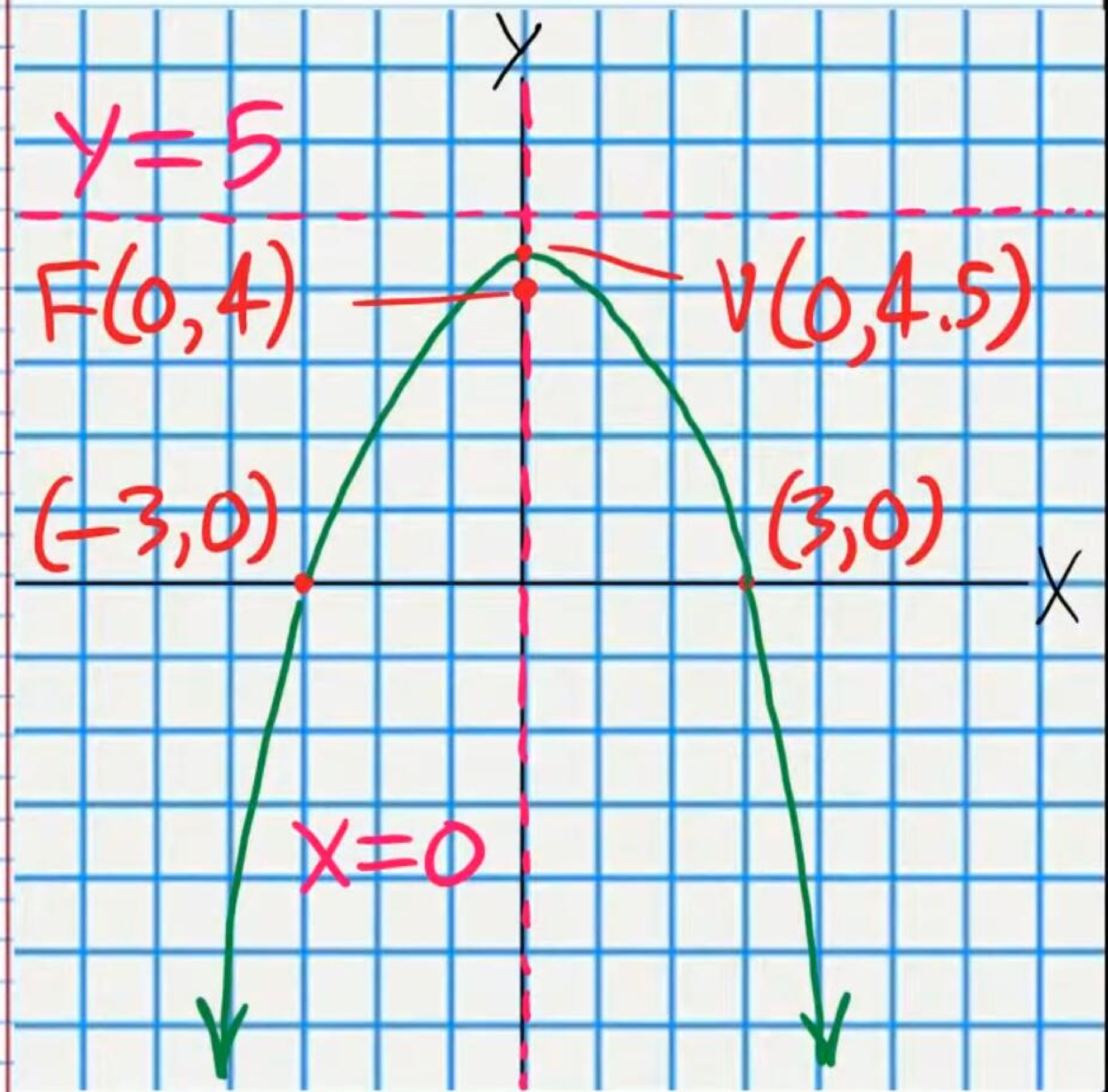
$y=5$

VI) Focus

$F(V_x,V_y+P)$

$F(0,4.5+(-\frac{1}{2}))$

$F(0,4)$



③ $-\frac{1}{4}(x+3)^2+3=y$

$a=-\frac{1}{4}$, $h=-3$, $k=3$

I) Vertex

$V_x=h=-3$, $V_y=k=3$

$V(-3,3)$

II) Line of Symmetry

$x=-3$

III) y-Intercept: $x=0$

$-\frac{1}{4}(0+3)^2+3=y$

$3/4=y$

$(0, 3/4)$

IV) x-Intercepts: $y=0$

$-\frac{1}{4}(x+3)^2+3=0$

$-\frac{1}{4}(x+3)^2=-3$

$-\frac{(x+3)^2}{4}=-\frac{3}{1}$

$-\frac{(x+3)^2}{4}=\frac{-12}{4}$

$-(x+3)^2=-12$

$(x+3)^2=12$

$\sqrt{(x+3)^2}=\pm\sqrt{12}$

$x+3=\pm 2\sqrt{3}$

$x=-3\pm 2\sqrt{3}$

$x=-3-2\sqrt{3}$ or $x=-3+2\sqrt{3}$

$(-6.464,0)$ & $(0.464,0)$

$(-3-2\sqrt{3},0)$ & $(-3+2\sqrt{3},0)$

V) Directrix

$P=\frac{1}{4a}=\frac{1}{4(-\frac{1}{4})}=-1$

$y=V_y-P$

$y=3-(-1)$

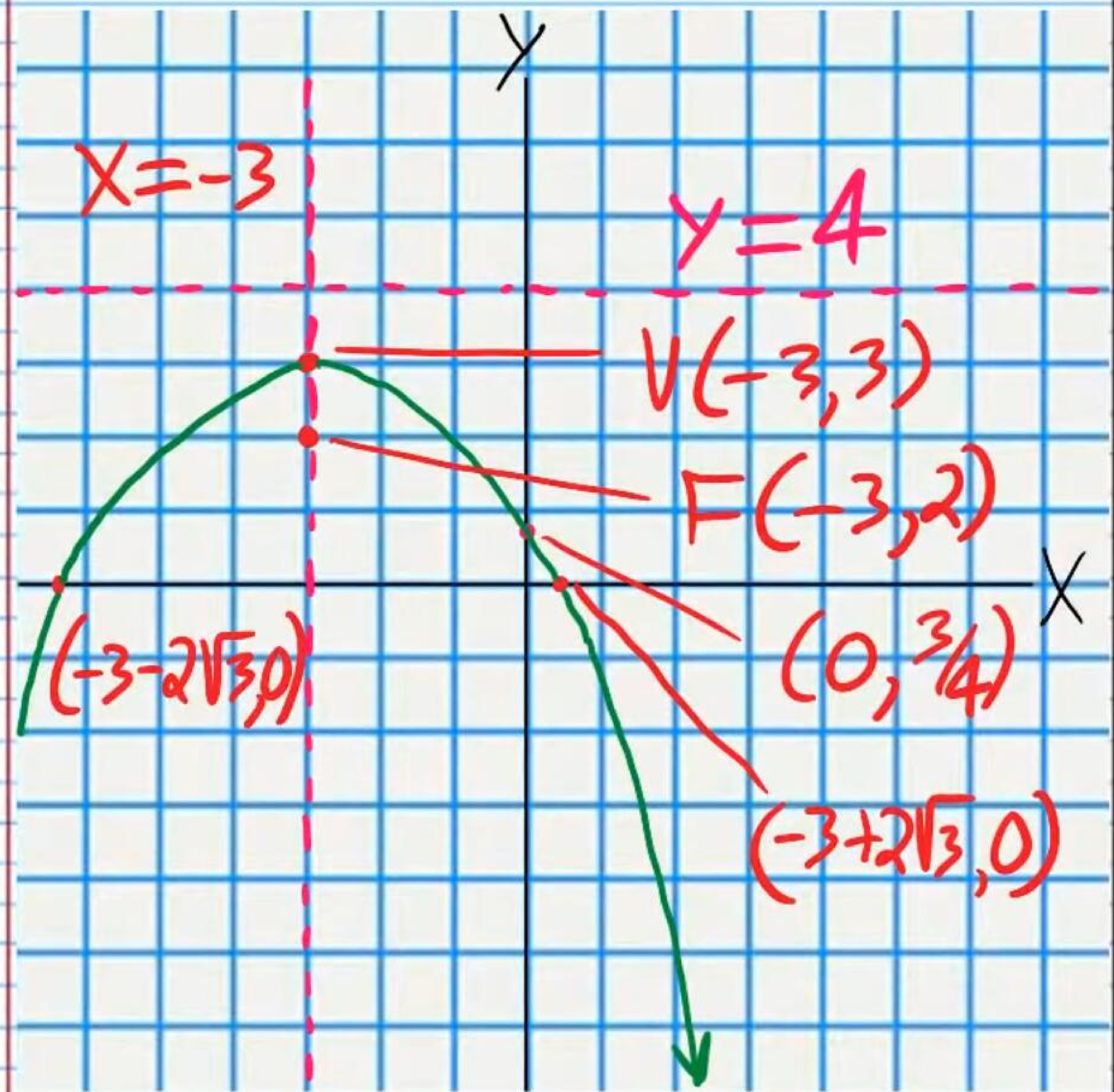
$y=4$

VI) Focus

$F(V_x, V_y+P)$

$F(-3, 3+(-1))$

$F(-3,2)$



④ $y=-2x^2+6x+3$

$a=-2,\ b=6,\ c=3$

I) Vertex

$V_x=-\frac{b}{2a}=-\frac{6}{2(-2)}=\frac{3}{2}$

$V_y=-2\left(\frac{3}{2}\right)^2+6\left(\frac{3}{2}\right)+3$

$V_y=-2\left(\frac{9}{4}\right)+9+3$

$V_y=-\frac{9}{2}+12=7\frac{1}{2}$

$V\left(1\frac{1}{2},7\frac{1}{2}\right)$

II) Line of Symmetry

$x=1\frac{1}{2}$

III) Y-Intercept: $x=0$

$(0,3)$

IV) X-Intercepts: $y=0$

$0=-2x^2+6x+3$

$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$x=\frac{-6\pm\sqrt{6^2-4(-2)(3)}}{2(-2)}$

$x=\frac{-6\pm\sqrt{6^2+24}}{-4}$

$x=\frac{-6\pm\sqrt{36+24}}{-4}$

$x=\frac{-6\pm\sqrt{60}}{-4}$

$x=\frac{-6\pm2\sqrt{15}}{-4}$

$x=\frac{-3\pm\sqrt{15}}{-2}$

$x=\frac{3\pm\sqrt{15}}{2}$

$x=\frac{3+\sqrt{15}}{2}$ or $x=\frac{3-\sqrt{15}}{2}$

$\left(\frac{3+\sqrt{15}}{2},0\right)$ & $\left(\frac{3-\sqrt{15}}{2},0\right)$

$(3.43,0)$ $(-0.43,0)$

V) Directrix

$P=\frac{1}{4a}=\frac{1}{4(-2)}=-\frac{1}{8}$

$y=V_y-P$

$y=7\frac{1}{2}-\left(-\frac{1}{8}\right)$

$y=7\frac{1}{2}+\frac{1}{8}$

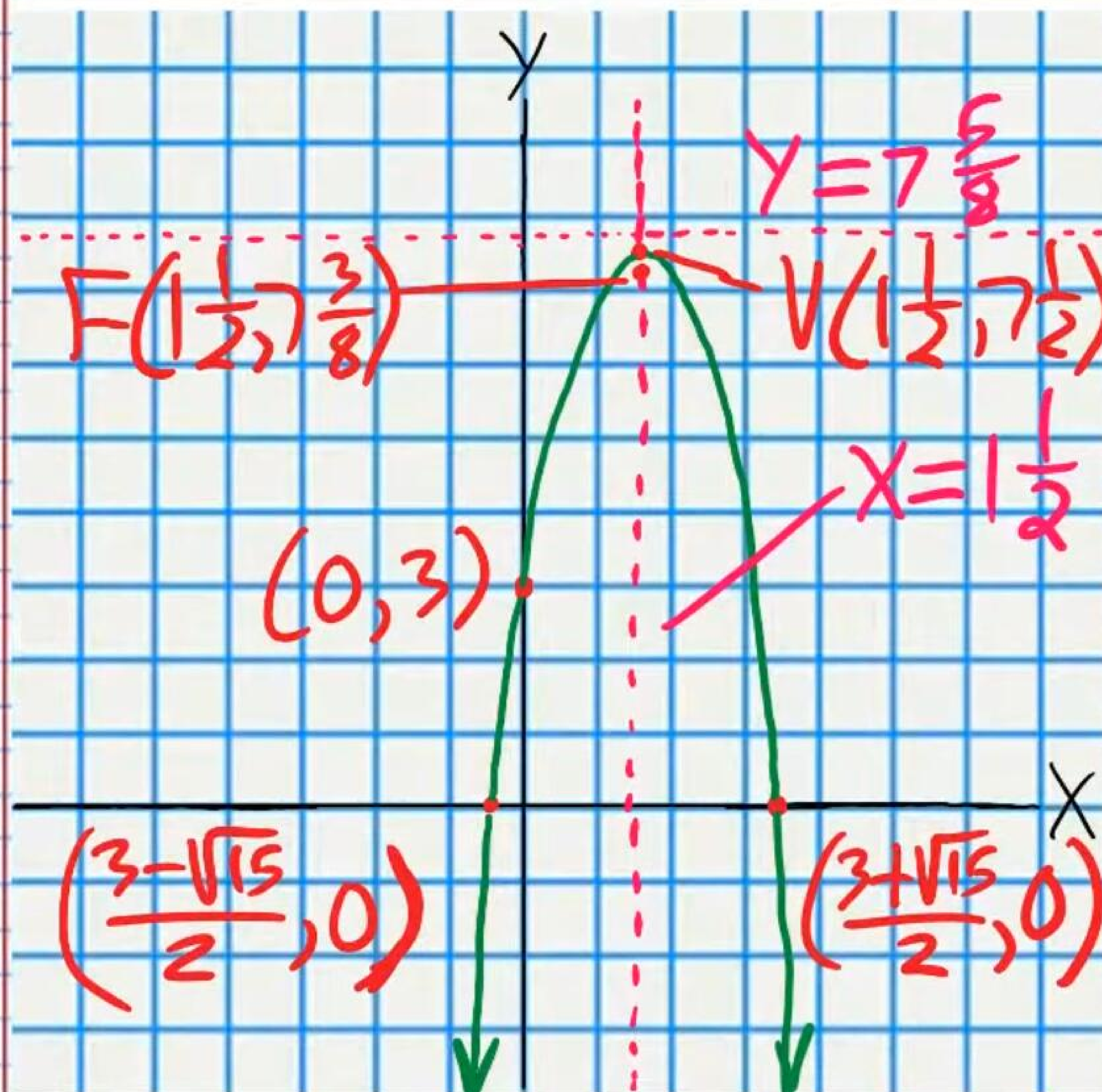
$y=7\frac{5}{8}$

VI) Focus

$F(V_x,V_y+P)$

$F\left(1\frac{1}{2},7\frac{1}{2}+\left(-\frac{1}{8}\right)\right)$

$F\left(1\frac{1}{2},7\frac{3}{8}\right)$



⑤ $y=2(x-1)^2+2$

$a=2,\ h=1,\ k=2$

I) Vertex

$V_x=h=1,\ V_y=k=2$

$V(1,2)$

II) Line of Symmetry

$x=1$

III) Y-Intercept: $x=0$

$y=2(0-1)^2+2=4$

$(0,4)$

IV) X-Intercepts: $y=0$

$0=2(x-1)^2+2$

$-2=2(x-1)^2$

$-1=(x-1)^2$

No X-intercepts

V) Directrix

$p=\frac{1}{4a}=\frac{1}{4(2)}=\frac{1}{8}$

$y=V_y-p$

$y=2-\frac{1}{8}$

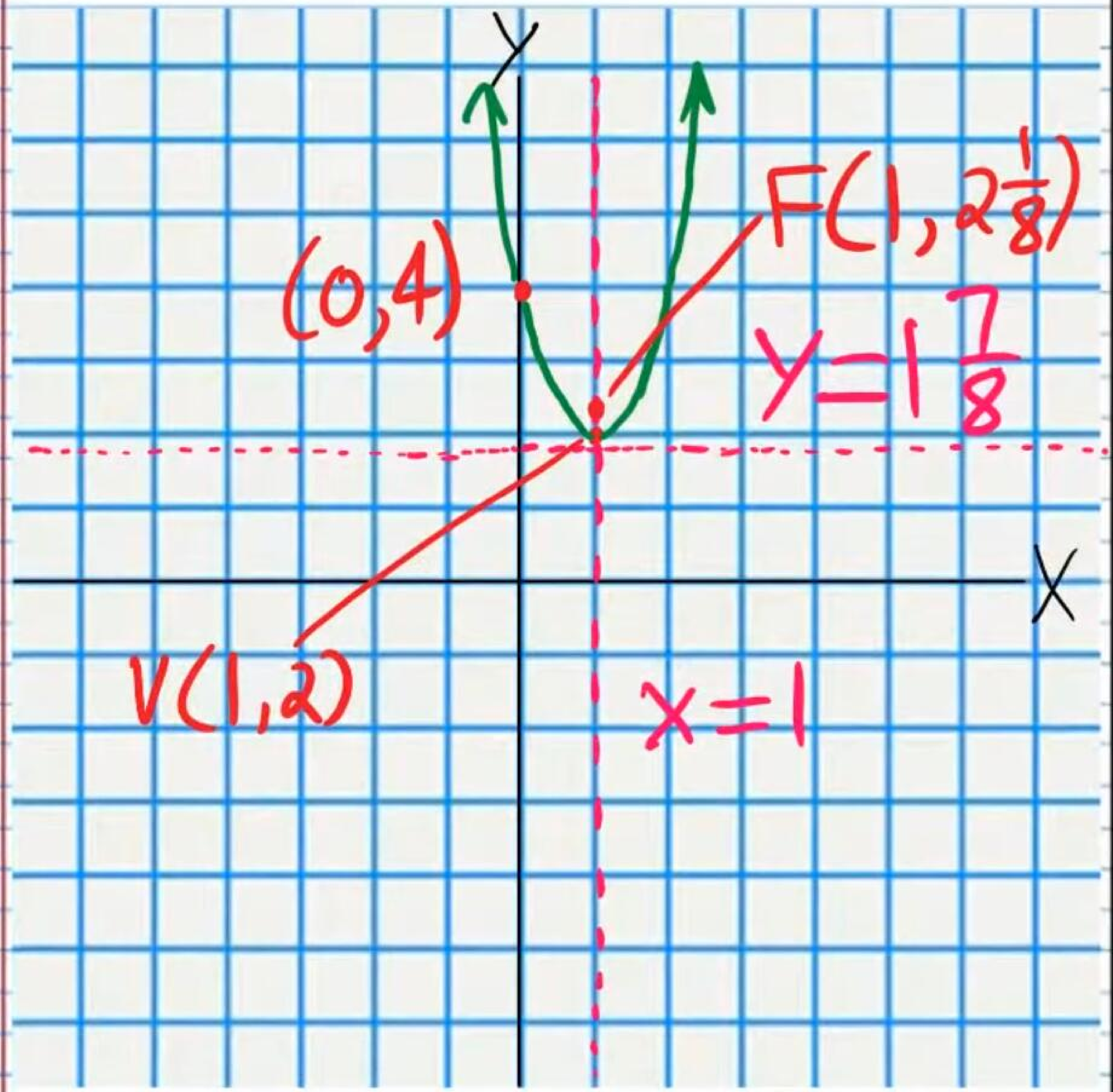
$y=1\frac{7}{8}$

VI) Focus

$F(V_x, V_y+p)$

$F(1, 2+\frac{1}{8})$

$F(1, 2\frac{1}{8})$



⑥ $y=(x-3)(x-4)$

$a=1, m=3, n=4$

I) Vertex

$V_x=\frac{m+n}{2}=\frac{3+4}{2}=3.5$

$V_y=(3.5-3)(3.5-4)$

$V_y=(0.5)(-0.5)$

$V_y=-\frac{1}{4}$

$V(3\frac{1}{2},-\frac{1}{4})$

II) Line of Symmetry

$x=3\frac{1}{2}$

III) Y-Intercept: $x=0$

$y=(0-3)(0-4)=12$

$(0,12)$

IV) X-Intercepts: $y=0$

$(3,0)$ & $(4,0)$

V) Directrix

$P=\frac{1}{4a}=\frac{1}{4(1)}=\frac{1}{4}$

$y=V_y-P$

$y=-\frac{1}{4}-\frac{1}{4}$

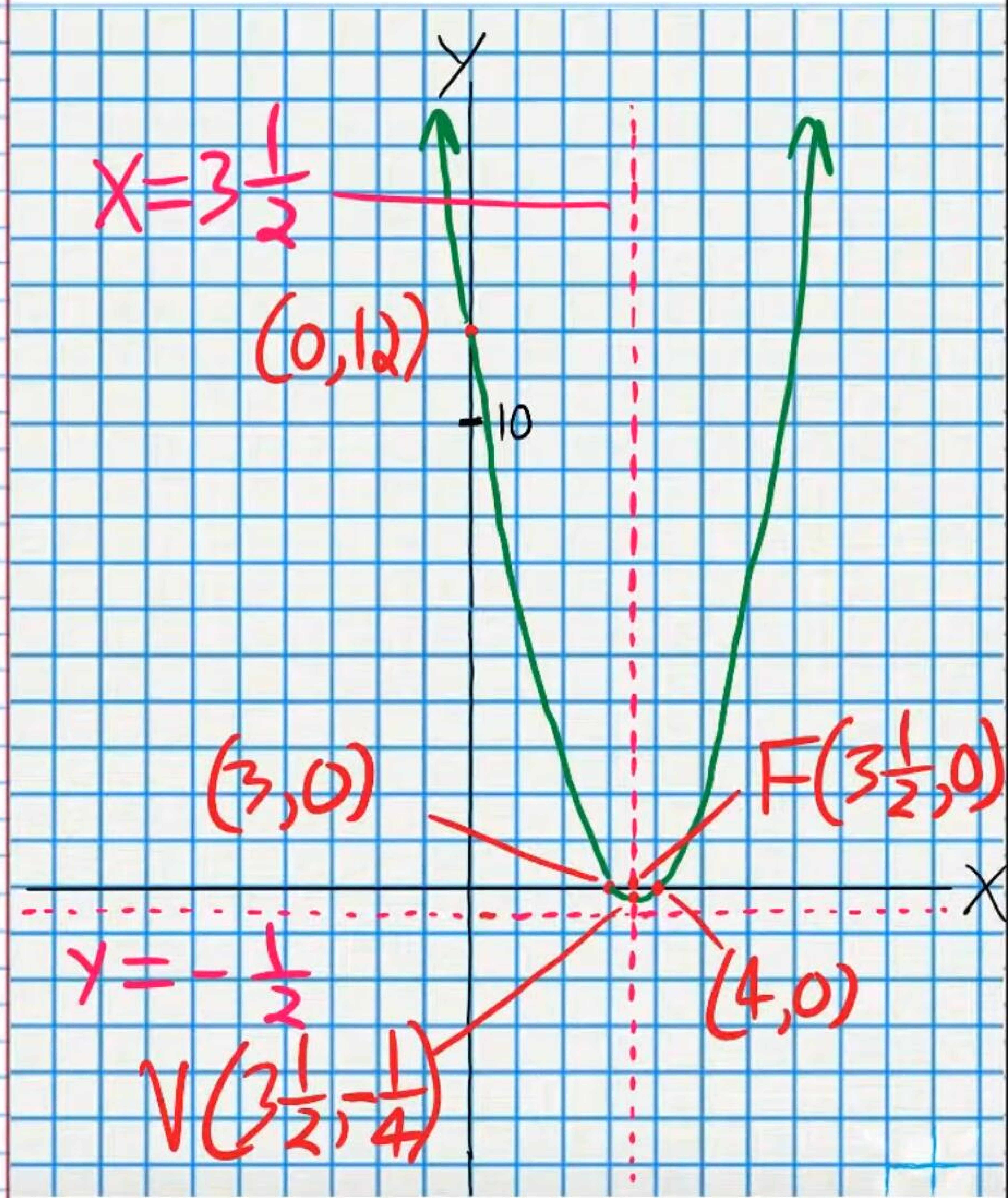
$y=-\frac{1}{2}$

VI) Focus

$F(V_x, V_y+P)$

$F(3\frac{1}{2},-\frac{1}{4}+\frac{1}{4})$

$F(3\frac{1}{2},0)$



⑦ $y=-x^2-4x+2$

$a=-1,\ b=-4,\ c=2$

I) Vertex

$V_x=-\frac{b}{2a}=-\frac{(-4)}{2(-1)}=-2$

$V_y=-(-2)^2-4(-2)+2$

$V_y=6$

$V(-2,6)$

II) Line of Symmetry

$x=-2$

III) Y-Intercept: $x=0$

$(0,2)$

IV) X-Intercepts: $y=0$

$0=-x^2-4x+2$

$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$x=\frac{-(-4)\pm\sqrt{(-4)^2-4(-1)(2)}}{2(-1)}$

$x=\frac{4\pm\sqrt{16+8}}{-2}$

$x=\frac{4\pm\sqrt{24}}{-2}$

$x=\frac{4\pm 2\sqrt{6}}{-2}$

$x=-2\pm\sqrt{6}$

$x=-2+\sqrt{6}$ or $x=-2-\sqrt{6}$

$(-2+\sqrt{6},0)$ & $(-2-\sqrt{6},0)$

$(0.45,0)$ & $(-4.45,0)$

V) Directrix

$P=\frac{1}{4a}=\frac{1}{4(-1)}=-\frac{1}{4}$

$y=V_y-P$

$y=6-\left(-\frac{1}{4}\right)$

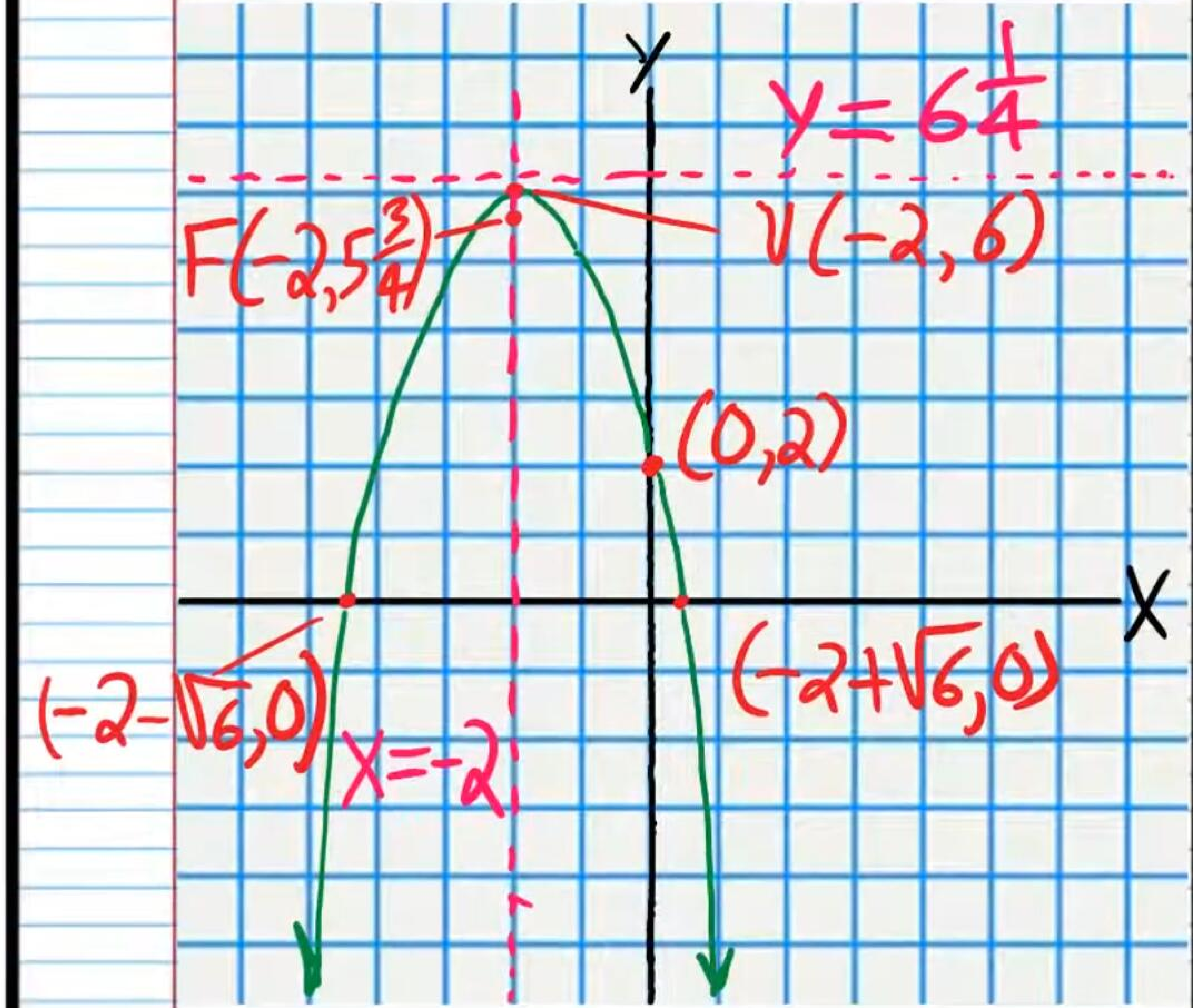
$y=6\frac{1}{4}$

VI) Focus

$F(V_x, V_y+P)$

$F\left(-2, 6+\left(-\frac{1}{4}\right)\right)$

$F\left(-2, 5\frac{3}{4}\right)$



⑧ $3x^2+14x+8=y$

$a=3,\ b=14,\ c=8$

I) Vertex

$V_x=-\frac{b}{2a}=-\frac{14}{2(3)}=-\frac{14}{6}=-\frac{7}{3}$

$V_y=3\left(-\frac{7}{3}\right)^2+14\left(-\frac{7}{3}\right)+8$

$V_y=3\left(\frac{49}{9}\right)-\frac{98}{3}+8$

$V_y=\frac{49}{3}-\frac{98}{3}+8$

$V_y=-\frac{49}{3}+8$

$V_y=-16\frac{1}{3}+8$

$V_y=-8\frac{1}{3}$

$V\left(-2\frac{1}{3},-8\frac{1}{3}\right)$

II) Line of Symmetry

$x=-2\frac{1}{3}$

III) Y-Intercept: $x=0$

$(0,8)$

IV) X-Intercepts: $y=0$

$3x^2+14x+8=0$

$(3x+2)(x+4)=0$

$3x+2=0$ or $x+4=0$

$x=-\frac{2}{3}$ or $x=-4$

$\left(-\frac{2}{3},0\right)$ & $(-4,0)$

V) Directrix

$p=\frac{1}{4a}=\frac{1}{4(3)}=\frac{1}{12}$

$y=V_y-p$

$y=-8\frac{1}{3}-\frac{1}{12}$

$y=-8\frac{5}{12}$

VI) Focus

$F(V_x,V_y+p)$

$F\left(-2\frac{1}{3},-8\frac{1}{3}+\frac{1}{12}\right)$

$F\left(-2\frac{1}{3},-8\frac{1}{4}\right)$